

# Zhenwu Xu

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## EDUCATION

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**Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences**

2020-2024      Ph.D.      *Physical Geography*

**School of Geography and Planning, Sun Yat-Sen University**

2017-2019      M.E.      *Hydraulic Engineering*

2013-2017      B.E.      *Physical Geography and Resources & Environment*

## WORK EXPERIENCES

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**Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences**

2024-present      Postdoctoral researcher

**Institute of Hydraulic Engineering and Water Resources Management, Technische Universität Wien**

2023-2024      Visiting student

## RESEARCH INTERESTS

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### *Global Change Ecohydrology*

- **Hydrological responses and resilience to disturbances (e.g., droughts, wildfires, deforestation)**
- **Process-based modeling for evapotranspiration, streamflow, and vegetation-water interactions**
- **Satellite–field data integration for global and regional assessment**

## PUBLICATIONS

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### *First author / corresponding author paper:*

- **Xu, Z.**, Zhang, Y., Kong, D., Ma, N., & Zhang, X. (2026). Extended global terrestrial evapotranspiration and gross primary production dataset from 1982 to near present. *Earth System Science Data*, 18, 5663–5695.
- **Xu, Z.**, Zhang, Y., Blöschl, G., & Piao, S. (2023). Mega forest fires intensify flood magnitudes in southeast Australia. *Geophysical Research Letters*, 50, e2023GL103812.
- **Xu, Z.**, Zhang, Y., Zhang, X., Ma, N., Tian, J., Kong, D., & Post, D. (2022). Bushfire-induced water balance changes detected by a modified paired catchment method. *Water Resources Research*, 58(11), e2021WR031013.
- **Xu, Z.**, Sun, L., Tang, G., Chen, X., Niu, X., Li, Y., & Yu, Y. (2022). Do model results vary under different routing algorithms based on a distributed ecohydrological model? *Journal of Hydrometeorology*, 23(12), 1913–1928.
- **Xu, Z.**, Tang, G., Jiang, T., Chen, X., Chen, T., & Niu, X. (2021). An automatic partition-based parallel algorithm for grid-based distributed hydrological models. *Environmental Modelling & Software*, 144, 105142.

### *Collaboration paper (selected):*

- Shao, X., Zhang, Y., Ma, N., Zhang, X., Tian, J., Xu, Z., & Liu, C. (2024). Drought-induced ecosystem

resistance and recovery observed at 118 flux tower stations across the globe. *Agricultural and Forest Meteorology*, 356, 110170.

- Lin, J., Bryan, B. A., Zhou, X., Lin, P., Do, H. X., et al. (2023). Making China's water data accessible, usable and shareable. *Nature Water*, 1, 328–335.
- Wang, L., Zhang, Y., Ma, N., Song, P., Tian, J., Zhang, X., & Xu, Z. (2023). Diverse responses of canopy conductance to heatwaves. *Agricultural and Forest Meteorology*, 335.
- Luan, J., Miao, P., Tian, X., Li, X., Ma, N., Xu, Z., et al. (2022). Separating the impact of check dams on runoff from climate and vegetation changes. *Journal of Hydrology*, 614, 128565.
- Chen, T., Tang, G., Yuan, Y., Guo, H., Xu, Z., Jiang, G., & Chen, X. (2020). Unraveling the relative impacts of climate change and human activities on grassland productivity in Central Asia over last three decades. *Science of The Total Environment*, 743, 140649.
- Tang, G., Li, S., Yang, M., Xu, Z., Liu, Y., & Gu, H. (2019). Streamflow response to snow regime shift associated with climate variability in four mountain watersheds in the US Great Basin. *Journal of Hydrology*, 573, 255–266.

## REVIEW ACTIVITIES

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### ORCID records:

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|-------------------------------------------------|---|
| • <i>Science Advances</i>                       | 1 |
| • <i>Water Resources Research</i>               | 2 |
| • <i>Journal of Hydrology</i>                   | 6 |
| • <i>Agricultural Water Management</i>          | 1 |
| • <i>Environmental Modelling &amp; Software</i> | 2 |

## GRANTS & AWARDS

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### Grants:

#### 2026 National Natural Science Foundation of China

##### Young Scientists Fund (Category C) (Grant No. 42601059)

*Mechanisms of Hydraulic Regulation on Carbon-Water Coupling in Typical Ecosystems of the Mu Us Sandy Land under Drought Stress*

#### 2025 China Postdoctoral Science Foundation

Fellowship Program (Grant No. GZC20250236)

General Program (Grant No.2025M770359)

*Evapotranspiration Responses and Recovery Mechanisms Following Forest Fire Disturbance in Southeastern China*

### Awards:

2023 Outstanding Student of the University of Chinese Academy of Sciences (UCAS)

## CONFERENCES

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### Session convener:

#### 2026 May General Assembly 2026 of the European Geosciences Union (EGU), Austria

“Ecohydrological responses to droughts in a changing environment” (1<sup>st</sup> convener)

2025 Nov. 26<sup>th</sup> International Congress on Modelling and Simulation (MODSIM), Australia  
 “Understanding hydrological extremes: trends, drivers and impacts” (3<sup>rd</sup> convener)

***Presentations:***

2026 May General Assembly 2026 of the European Geosciences Union (EGU), Austria  
 2025 Nov. 26<sup>th</sup> International Congress on Modelling and Simulation (MODSIM), Australia  
 2024 Apr. General Assembly 2024 of the European Geosciences Union (EGU), Austria  
 2023 Jul. 25<sup>th</sup> International Congress on Modelling and Simulation (MODSIM), Australia  
 2021 Dec. 24<sup>th</sup> International Congress on Modelling and Simulation (MODSIM), Australia  
 2018 Nov. 1<sup>st</sup> Symposium of Space Earth Science, Sanya, China

## TEACHING EXPERIENCES

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|-------------|------|-------------------------------|--------------------|
| 2025 Fall   | UCAS | Modern Hydrology              | Teaching Assistant |
| 2024 Fall   | UCAS | Modern Hydrology              | Teaching Assistant |
| 2018 Fall   | SYSU | Terrestrial Ecosystem Ecology | Teaching Assistant |
| 2018 Spring | SYSU | C++ programming               | Teaching Assistant |

## TECHNICAL SKILLS

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- **Model** **PML-V2 (Diagnostic ET and GPP model)**  
**CHESS (Distributed ecohydrological model)**  
 HBV, GR4J, Xinanjiang, etc. (Lumped hydrological models)
- **Coding** R C++ Fortran Linux JavaScript (Google Earth Engine)
- **Language** English Chinese (Mandarin, Cantonese, Hokkien, Hakka)